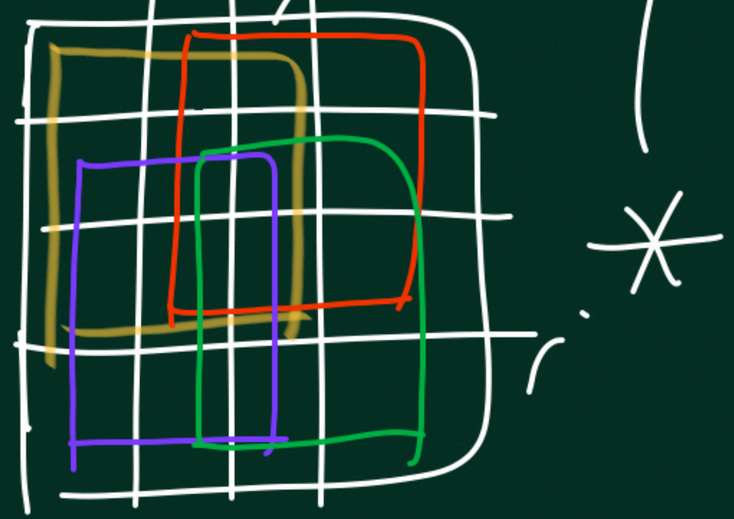
Convolutional Neural Nets

Convolution –

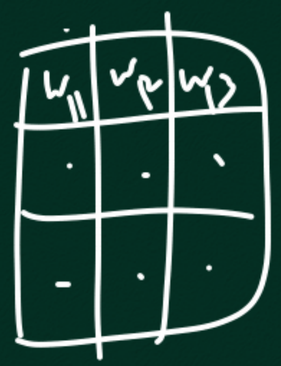
receptive field

Convolution.



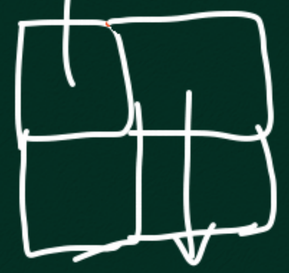
Input image

row x col



filter/ kernel.

$$z_{11} = \sum_{r=1}^{3}\sum_{c=1}^{3} w_{rc}\, p_{\sigma_r(r,1)\,\sigma_c(c,1)}$$



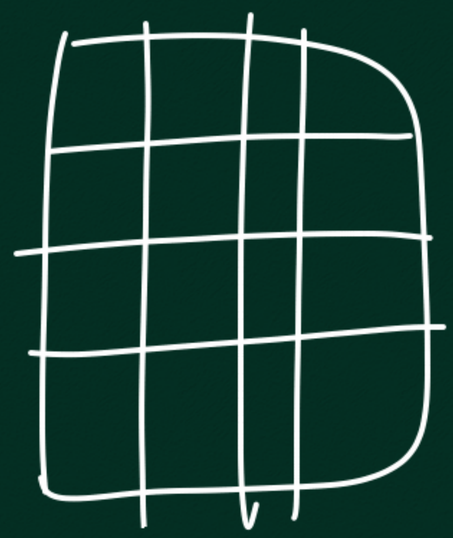
$$z_{12} = \sum\sum w_{rc}\, p_{\sigma_r(r,1)\,\sigma_c(c,2)}$$

$$z_{ij} = \sum_{r=1}^{h} \sum_{c=1}^{w} w_{rc} \beta \sigma_r(r,i) \sigma_c(c,j)$$

$$a_{ij} = \phi(z_{ij})$$

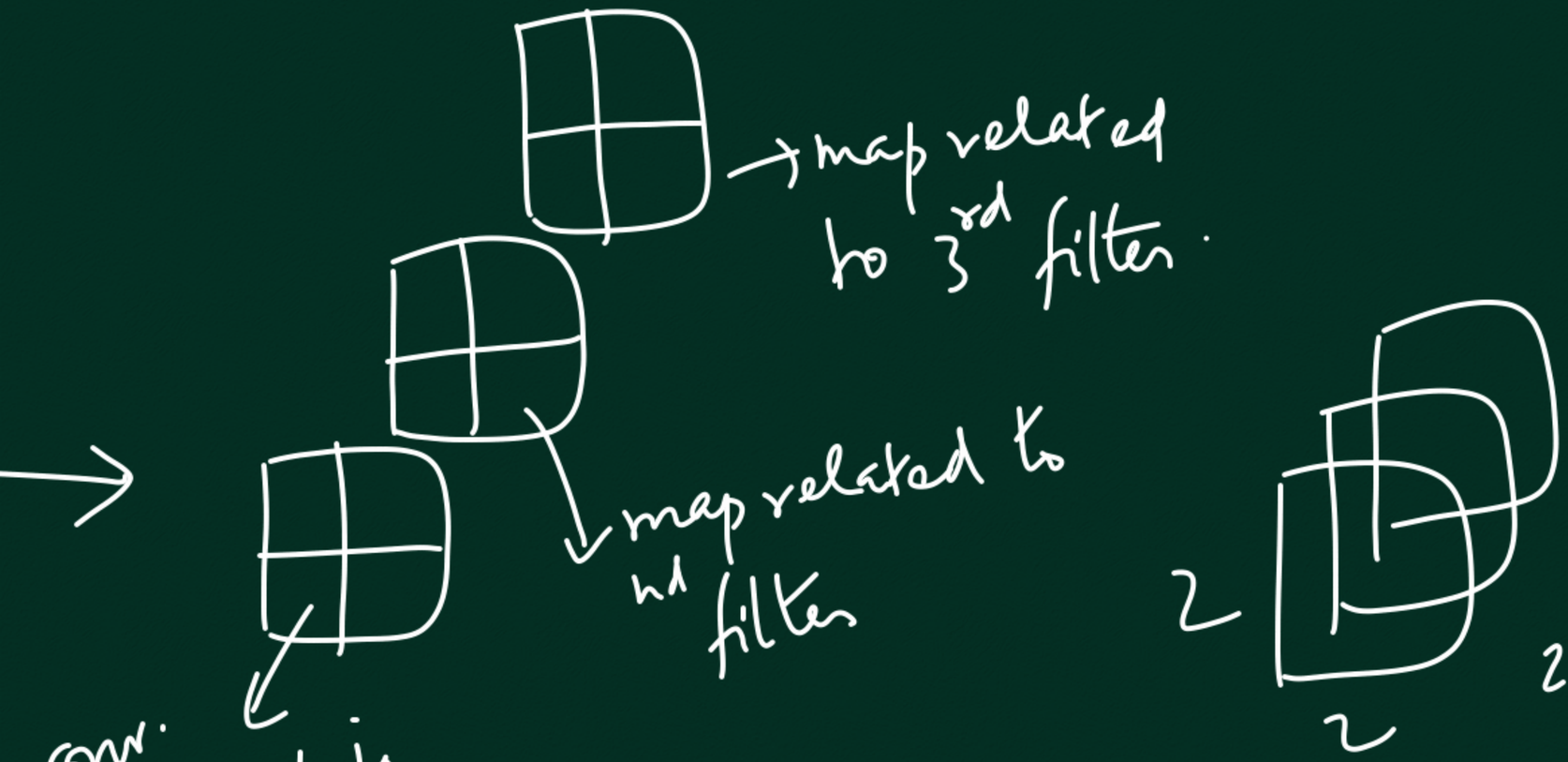






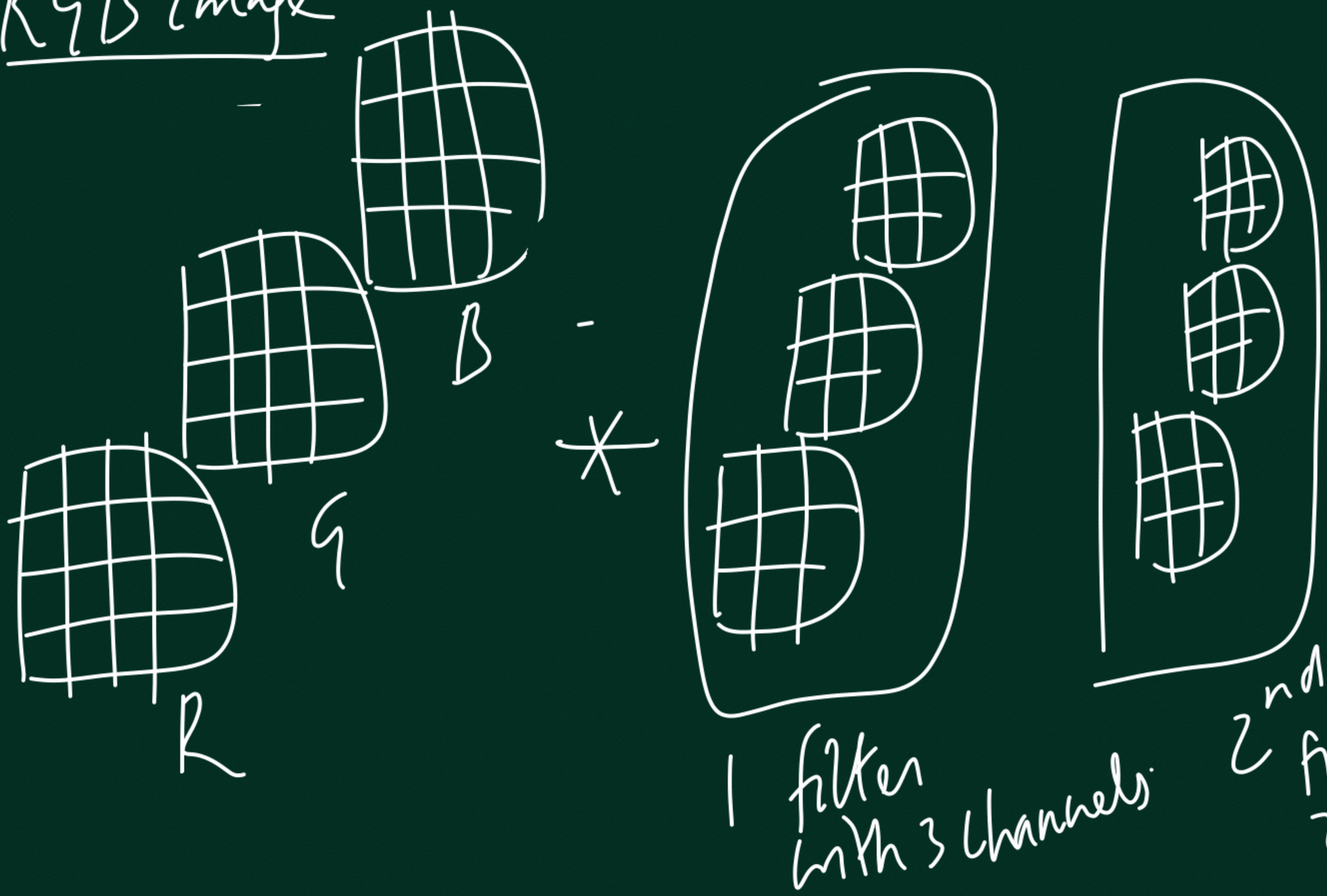
4x4





Num of channels in convolved feature map } = Num of filters used for conv operation

RGB image



$$z_{ij} = \sum\sum w_{rc} p_{f(r,i)\,f(j,c)}$$

3 channel input

1 filter of 3 channels

1 channel output

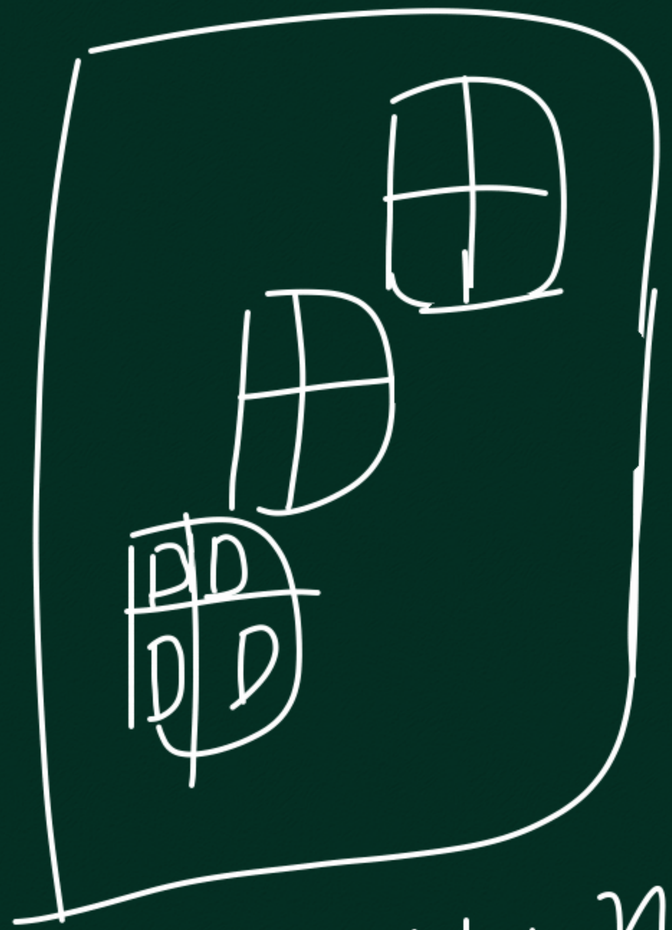
$$z_{ij}^{k} = \sum_{r} \sum_{c} w_{rc}^{k} \; p_{f(r,i) f(j,c)}$$

k: kernel / channel index.

r: row

c: col.





$*$



$$z_{ij}^{f} = \sum_{k=1}^{\text{num of channels}} \sum_{r} \sum_{c} w_{rc}^{k,f} \, \beta^{k} \, f(r,i) f(c,j)$$

r ↓ row c ↓ col.

$[X]$



a map with

3 channels each of size 2×2

$\phi(Wx)$

$W \rightarrow 784 \times 100$

784 X 1

100 hidden layer

784 28 28

3X3

2

3x3

map.

4096

x4096

Stride

2x2

map

num of filters

Kernel width

stride

pad

padding

3x3

*

2x2

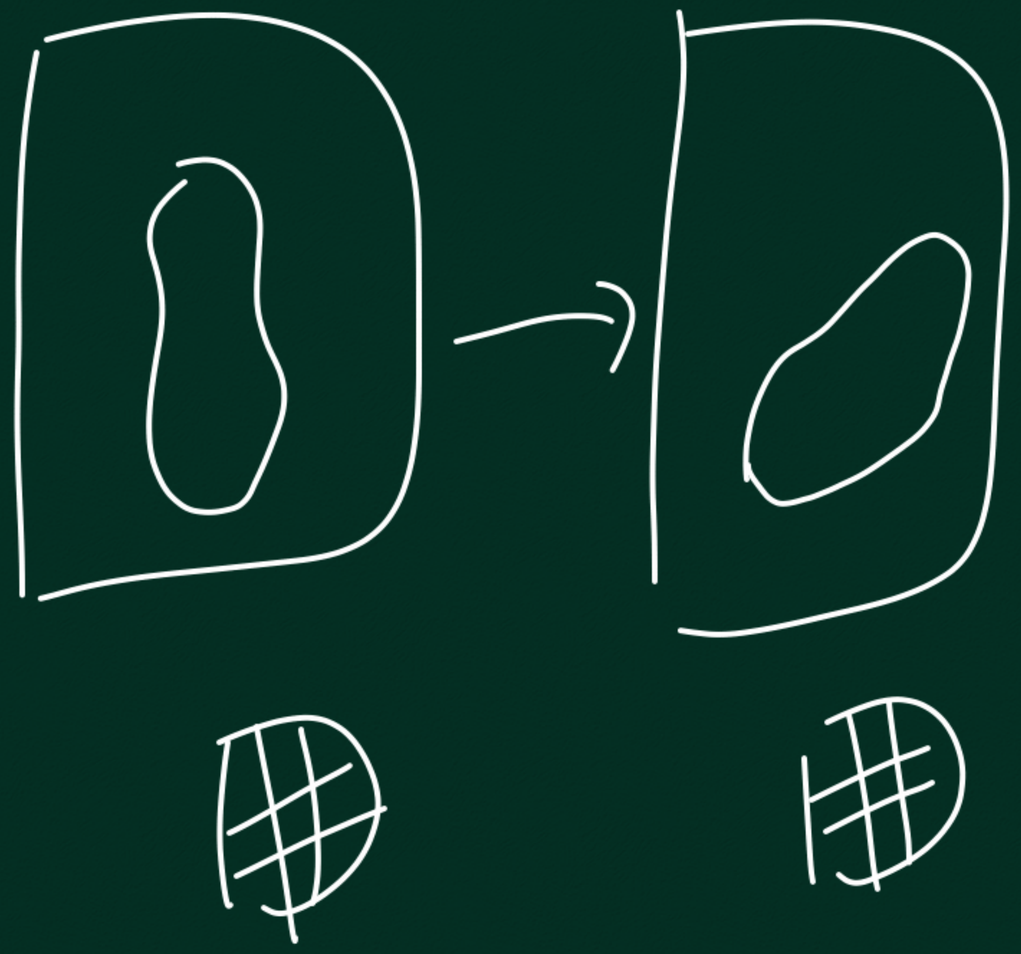
→

2X

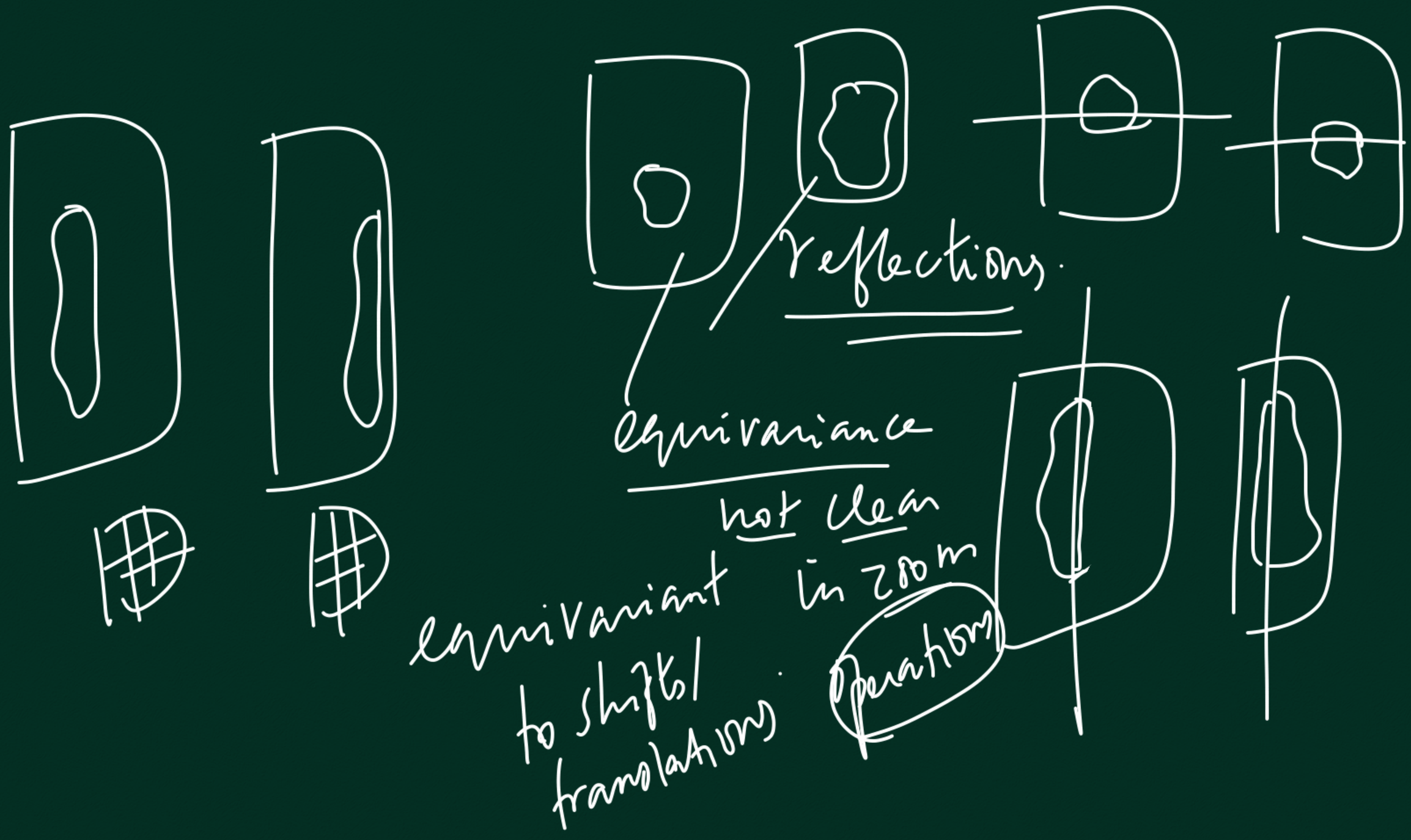
Conv map

3x3

output



equivariance
to rotation

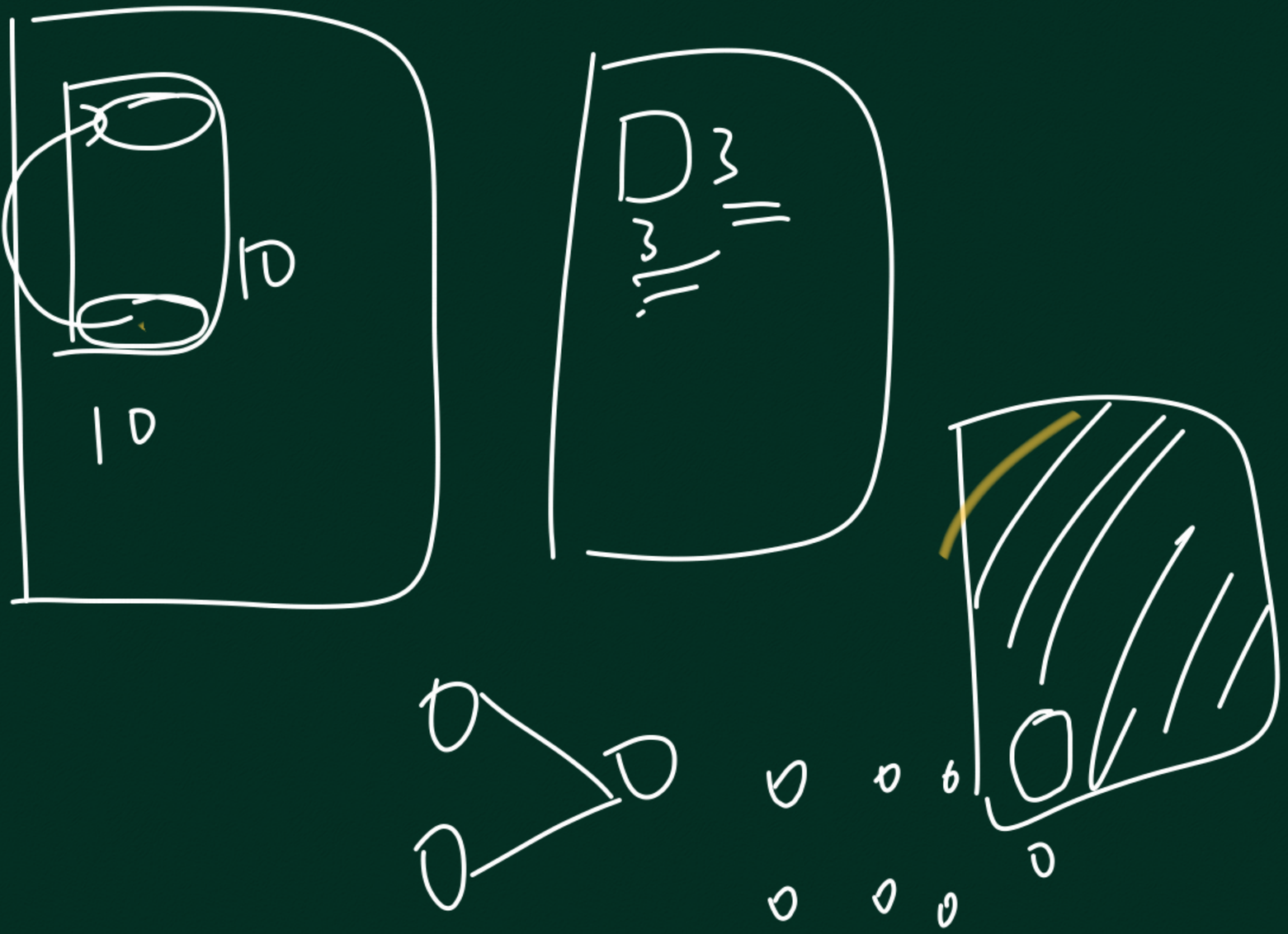

reflections.
equivariance
not clear
equivariant in zoom
to shifts/
translations
operations

Input C_1 C_2 C_3 C_K O/p

$\phi(W x)$ $\phi(W a^1)$ $\phi(W a^2)$ . . .

W W

W W



Pooling

max pooling



dim reduction

max Intensity



mean int

avg intensity

wt.avg

2x2 pooling operation.

